THE INDUCED MORPHISM ON FIBRES IN A CARTESIAN SQUARE WITH ETALE VERTICAL ARROWS IS ETALE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Etale morphisms are stable under base change and are open immersions locally on the source. Standard, used at proof lines 8199-8200. Gives that the base changed vertical arrow is etale.
- The fibre of an etale morphism over a point is a disjoint union of spectra of finite separable field extensions. Morphisms, Lemma morphisms-lemma-etale-over-field, invoked at proof lines 8201-8204. Identifies the chosen point of the upper right scheme with one of the summands, making the fibre over it open and closed in the base changed fibre.
- The composite of an open immersion with an etale morphism is etale. Morphisms, Section morphisms-section-etale, invoked at proof lines 8205-8208. Upgrades the open and closed embedding of fibres to the asserted etale morphism.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 02GH: sources/stacks-chow/tags.txt, source lines 3228–3228. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Consider a commutative diagram of morphisms of schemes*

$$\begin{array}{ccc} U' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ U & \longrightarrow & V \end{array}$$

with étale vertical arrows and a point $v' \in V'$ mapping to $v \in V$. Then the morphism of fibres $U'_{v'} \to U_v$ is étale.

Proof. Note that $U'_v \to U_v$ is étale as a base change of the étale morphism $U' \to U$. The scheme U'_v is a scheme over V'_v. By Morphisms, Lemma 02GL the scheme V'_v is a disjoint union of spectra of finite separable field extensions of $\kappa(v)$. One of these is $v' = \mathrm{Spec}(\kappa(v'))$. Hence $U'_{v'}$ is an open and closed subscheme of U'_v and it follows that $U'_{v'} \to U'_v \to U_v$ is étale (as a composition of an open immersion and an étale morphism, see Morphisms, Section 02GH). □